Week 7:

1. In Approximate Differential Privacy, what does δ represent?
    a. The strict upper bound of the probability of the mechanism M.
    b. The estimated noise added to the Laplace Mechanism.
    c. The average difference between all the data points in the datasets X and X'.
    d. A leeway for small-probability events where ε-differential privacy may fail.

    Explanation: Approximate Differential Privacy [6:50-9:26] - δ is a small leeway for low probability events where ε-differential privacy property may not hold true.

2. Compared to the Laplacian mechanism, how does the Gaussian mechanism scale the amount of noise added when releasing D different outputs?
    a. It adds noise linearly proportional to D.
    b. It adds noise proportional to the square root of D.
    c. It adds noise exponentially proportional to D.
    d. It adds the same amount of noise.

    Explanation: Approximate Differential Privacy [18:00-20:58] - The Guassian Mechanism requires quadratically less amount of noise to become in terms of number of output that you're releasing.

3. Which of the following are properties of Approximate Differential Privacy? **(Select all that apply.)**
    a. It satisfies the property of post-processing, meaning any data-independent computation applied to the output will not make it less (ε, δ)-differentially private.
    b. It guarantees privacy even when δ is significantly larger than $1/n$ (where n is the number of individuals in a dataset).
    c. It satisfies composition, meaning if you have k mechanisms that are all (ε, δ) -differentially private, then the combined mechanism is $(k\varepsilon, k\delta)$-differentially private.
    d. It guarantees that the output distributions of the algorithm are completely identical for any two neighboring datasets.

    Explanation: Approximate Differential Privacy [23:40-34:53] - Approximate differential privacy has two properties: post-processing and composition.

4. According to the lecture, what is the primary advantage of using the Gaussian Mechanism?
    a. Zero chance of privacy leakage.
    b. Faster computational time for calculating probabilities.
    c. Better utility due to adding quadratically less noise.
    d. Eliminates the need for a trusted curator.

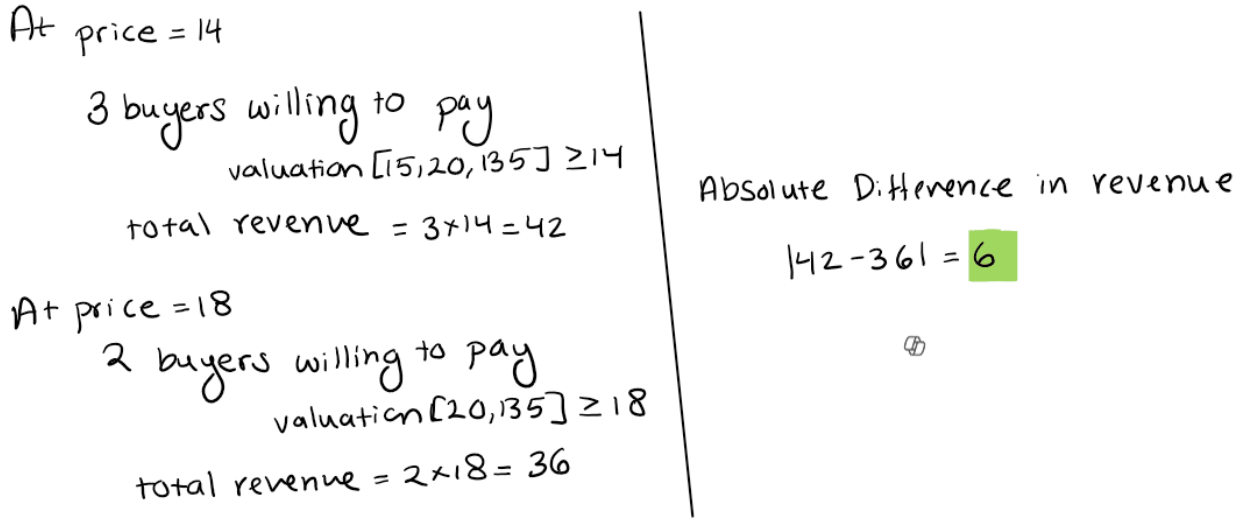
    Explanation: Approximate Differential Privacy [14:45-22:55] - Gaussian mechanism gains better utility with quadratically less amount of noise.

5. There are three buyers whose valuations are 15, 20, and 135, respectively. What is the absolute difference in revenue when the price of a digital good changes from 14 to 18?

(Round your answer to the nearest whole number.)

Answer (numerical): 6

Explanation: Exponential Mechanism



6. How does the Exponential Mechanism select a price $h \in H$?
   a. With probability proportional to the inverse of the price.
   b. With probability proportional to $e^{\frac{\varepsilon}{2\delta}S(x,h)}$.
   c. With probability proportional to $e^{\frac{\delta}{2\varepsilon}S(x,h)}$.
   d. Uniformly at random from all prices below the maximum valuation.

   Explanation: Exponential Mechanism [14:11-15:00] - Select $h \in H$ with probability proportional to $e^{\frac{\varepsilon}{2\delta}S(x,h)}$.

7. What privacy guarantee does the Exponential Mechanism provide for buyers' valuations?
   a. (ε, δ)-Differential Privacy
   b. ε-Differential Privacy
   c. K-Anonymity
   d. Concentrated Differential Privacy

   Explanation: Exponential Mechanism [15:53-18:15] - Exponential mechanism is Epsilon differentially private for the given valuations.

8. What is the utility guarantee provided by the Exponential Mechanism?
   a. It guarantees the absolute highest possible revenue every time.
   b. It does not guarantee positive revenue every time.
   c. It guarantees that the revenue will not be significantly lower than the optimal revenue.
   d. The guarantee will exactly match the median valuation.

Explanation: Exponential Mechanism [18:15-22:35] - It guarantees that the revenue will not be significantly lower than the optimal revenue.

9. In Statistical Parity, for a protected group C and a classifier M, what is the parity value for an ideal classifier?
    a. -1
    b. 0
    c. 0.5
    d. 1

    Explanation: Fairness in Machine Learning [47:03-50:41] - An ideal classifier has a parity equal to 0.

10. According to the data comparing White and African American defendants in risk assessments, which of the following statements are true? **(Select all that apply.)**
    a. White defendants were labeled lower risk yet did re-offend at a higher rate than African American defendants.
    b. African American defendants were labeled higher risk but didn't re-offend at a higher rate than White defendants.
    c. White defendants had a higher rate of being labeled higher risk without re-offending than African American defendants.
    d. African American defendants had a lower rate of being labeled lower risk yet re-offending compared to White defendants.

    Explanation: Fairness in Machine Learning [19:42-21:38] - According to the table:
    - A higher percentage of African Americans were labelled as higher risk but didn't re-offend
    - A higher percentage of White people were labelled as low risk but did re-offend.
    - A lower percentage of African Americans were labelled as lower risk yet re-offended compared to White people.

11. Based on the analysis of biases in word embeddings from Wikipedia and Social Media, which of the following professions was associated with sexist prejudice towards men?
    a. Nurse
    b. Teacher
    c. Pharmacist
    d. Cameraman

    Explanation: Fairness in Machine Learning [24:44-28:10] - According to the lecture, "Cameraman" is a profession that is associated with sexist prejudice towards men.

12. The goal of Fair PCA is to find a PCA solution $U = [U_A, U_B]$ such that the reconstruction losses of groups A and B (where A is the protected group) are:
    a. Equal
    b. Unequal
    c. The protected group has a lower reconstruction loss
    d. The protected group has a higher reconstruction loss

Explanation: Fairness in Machine Learning [40:05-44:56] - At optimality, Fair PCA minimizes the maximum loss between the groups, which mathematically forces their average reconstruction errors to be equal.